

Breastfeeding and Once-Daily Small-Volume Formula Supplementation to Prevent Infant Growth Impairment

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abstract

BACKGROUND: Randomized controlled trials in Guinea-Bissau and Uganda have revealed that the intensive promotion of exclusive breastfeeding (EBF) impairs growth in early infancy. When newborn growth is impaired, small amounts of formula may be combined with breastfeeding to promote growth.

METHODS: To determine if breastfeeding combined with once-daily formula supplementation improves growth among at-risk newborns, we conducted a pilot randomized controlled trial in Bissau, Guinea-Bissau and Kampala, Uganda. We randomly assigned 324 healthy breastfeeding newborns who weighed 2000 g to 2499 g at birth or <2600 g at 4 days old to once-daily formula feeding through 30 days as a supplement to frequent breastfeeding followed by EBF from 31 days through 6 months, or to EBF through 6 months. The primary outcome was weight-for-age z score (WAZ) at 30 days. Other outcomes included weight-for-length z score (WLZ), length-for-age z score (LAZ), breastfeeding cessation, adverse events, and serious adverse events through 180 days.

RESULTS: Daily formula consumption in the intervention group was 31.9 ± 11.8 mL. The random assignment did not impact WAZ, WLZ, LAZ, breastfeeding cessation, adverse events, or serious adverse events through 180 days. In the intervention and control groups, 19 (12%) and 35 (21%) infants, respectively, reported nonformula supplementation in the first 30 days ($P = .02$).

CONCLUSIONS: Once-daily formula supplementation for 30 days was well-tolerated, but the small volume consumed did not alter growth through 180 days of age. Further research would be required to determine if larger formula volumes, longer duration of treatment, or more frequent feeding are effective at increasing growth for this at-risk population.

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WHAT'S KNOWN ON THIS SUBJECT: In Guinea-Bissau and Uganda, randomized controlled trials have revealed that the intensive promotion of exclusive breastfeeding impairs growth in early infancy. When newborn growth is impaired in high-income countries, small amounts of infant formula may be combined with breastfeeding to promote growth.

WHAT THIS STUDY ADDS: Among at-risk breastfeeding newborns in Guinea-Bissau and Uganda, once-daily formula supplementation added to frequent breastfeeding for 30 days was well-tolerated; however, the small volume of formula consumed did not alter growth through 180 days of age.

To cite: Flaherman VJ, Murungi J, Bale C, et al. Breastfeeding and Once-Daily Small-Volume Formula Supplementation to Prevent Infant Growth Impairment. *Pediatrics*. 2024;153(1):e2023062228

Growth impairment is highly prevalent among young children in Uganda and Guinea-Bissau. In Uganda, Demographic and Health Survey data reveal that 12% of infants <6 months of age meet the criteria for stunting, whereas 8% meet the criteria for wasting¹; in Guinea-Bissau, the prevalence of stunting among children <5 years of age is 28%.² The impact of these growth deficits is profound. Stunting hinders brain development and increases the risk of cognitive impairment^{3–7}; wasting weakens immunity, leaving children susceptible to infection and mortality.^{8,9} The detrimental impact of growth impairment accounts in part for the high infant mortality rates in Guinea-Bissau and Uganda, which exceed 42 per 1000^{1,10} and place a tremendous burden on families and communities, public health and educational systems, and local, regional, and national economies.^{11,12} Thus, reducing the prevalence of growth impairment is a critically important global health priority.^{13–15}

The World Health Organization (WHO) Child Growth Standards reveal that the first 30 days after birth are a pivotal period during which newborns typically gain 1% of their body weight each day, with especially rapid development of brain tissue.^{16–19} Exclusive breastfeeding (EBF) is recommended during this time, and for the first 6 months by the WHO, and is a cornerstone of public health efforts to improve infant outcomes in sub-Saharan Africa.^{15,20} In observational studies, the receipt of EBF for the first 6 months is associated with optimal outcomes both in high-income countries (HIC) and in low- and middle-income countries (LMIC).^{21,22} Observational studies also reveal that infants who receive EBF for 6 months have lower risks of infectious diseases, diarrhea, and mortality and show better cognitive function than supplemented infants.^{23,24}

Few randomized controlled trials have been conducted comparing EBF to other feeding interventions. The landmark Promotion of Breastfeeding Intervention (PROBIT) cluster randomized controlled trial was conducted in Belarus in 1996 and 1997 and found a beneficial impact of intensive promotion of EBF on both growth and gastroenteritis; PROBIT results informed the development of current WHO feeding guidelines.²⁵ Although PROBIT revealed a beneficial effect of EBF,²⁶ 2 randomized controlled trials conducted subsequently in sub-Saharan Africa revealed results that strongly contrast with PROBIT. The PROMISE-EBF cluster-randomized controlled trial revealed that, in Uganda, the intensive promotion of EBF slowed growth through 24 weeks of age and increased the risk of growth impairment at age 5 years without reducing morbidity or mortality.^{27,28} Similarly, the Bandim Health Project randomized controlled trial conducted in Guinea-Bissau revealed that the intensive promotion of EBF decreased infant weight at 4 to 6 months of age without reducing morbidity or mortality.²⁹ Taken together, the experimental evidence, therefore, suggests that in

Guinea-Bissau and Uganda, interventions other than EBF may be needed to prevent growth impairment.

In HIC, the trajectory of weight gain for breastfed newborns supplemented with formula approximates WHO Child Growth Standards more closely than does the trajectory of weight gain for exclusively breastfed newborns.^{30,31} It is possible that supplementary formula might have a similar impact on weight gain for breastfeeding newborns in Guinea-Bissau and Uganda. However, formula supplementation is not recommended in Guinea-Bissau and Uganda because of the observational evidence linking supplementation to morbidity, mortality, and breastfeeding cessation.^{32–35} One potential strategy for optimizing newborn growth and avoiding breastfeeding cessation and potential morbidity and mortality is early small-volume formula supplementation (ESVFS), a technique tested in HIC and found to allow the transition back to EBF after the rapid growth demands of the newborn period have passed.^{36–38} The applicability of HIC findings regarding ESVFS to settings in Guinea-Bissau and Uganda is unknown.

If the temporary use of ESVFS in the newborn period increases weight among breastfeeding at-risk infants in Guinea-Bissau and Uganda without a deleterious impact on adverse events (AEs), mortality, or subsequent breastfeeding cessation, ESVFS could potentially improve growth and prevent stunting and wasting in this vulnerable population. To determine the effect of ESVFS on growth and other health metrics among low birth weight and at-risk breastfeeding infants in Guinea-Bissau and Uganda, we conducted the Preventing Infant Malnutrition with Early Supplementation (PRIMES) randomized controlled pilot trial.

METHODS

Trial Design, Participants, and Setting

PRIMES eligibility criteria and trial methods have been published.³⁹ In Bissau, Guinea-Bissau, and Kampala, Uganda, infants eligible for PRIMES were healthy singleton breastfeeding newborns. Key eligibility criteria included a weight of 2000 g to 2885 g at <6 hours after birth and negative maternal HIV testing; eligible infants were randomly assigned if their weight was <2500 g at <6 hours after birth (Day 0) or <2600 on Day 4. This study was approved by the Guinea-Bissau Comité Nacional de Ética na Saúde (Guinea-Bissau National Committee on Ethics and Health, 075/CNES/INASA/2020), the Higher Degrees, Research and Ethics Committee of Makerere University (871), the Uganda National Council of Science and Technology (HS1226ES), and the University of California San Francisco Institutional Review Board (19-29405).

Recruitment and Randomization

Recruitment, enrollment, and randomization occurred from February to December 2021. Local clinicians informed

mothers about the study, asked them about potential interest in study participation, and referred those who expressed interest to study staff, who then described the study in detail, including all study procedures, the randomization process, and the voluntary nature of participation, and encouraged mothers to ask questions. Structured items were then used to ascertain maternal comprehension of the nature of randomization and the ongoing voluntary nature of participation, and written informed consent was then obtained by signature or thumbprint. After obtaining informed consent, trained study staff obtained treatment assignment generated by the online software application, randomize.net, with randomly permuted blocks of 4 and 6 infants stratified by site (Guinea-Bissau or Uganda) and weight (2000–2249 g on Day 0, 2250–2499 g on Day 0, 2250–2424 g on Day 4, and 2425–2599 g on Day 4).

Interventions

For enrolled infants randomly assigned to the ESVFS (intervention) group, trained study staff provided breastfeeding support and taught the mothers to breastfeed the infant frequently on demand and at least 8 times per day, and to follow 1 completed breastfeeding per day with the offer of a premixed, commercially sterile, single-use 59 mL bottle of Similac Pro Advance (Abbott Nutrition, Columbus, OH), allowing the infant to first consume breast milk until the infant discontinued breastfeeding and then consume the supplement until the infant discontinued feeding. Powdered formula was not used, and no formula was mixed by the study staff or participants. Trained study staff used a structured tool to teach mothers to assemble each ESVFS feeding in a hygienic manner from a commercially sterile, single-use container and a single-use sterile nipple and then observed 4 daily feedings of breastfeeding followed by ESVFS feeding, providing feedback on breastfeeding, ESVFS assembly, and ESVFS feeding technique when needed. After these directly observed feedings, trained study staff delivered two 59 mL formula bottles and 2 single-use sterile nipples to infants in the ESVFS group every other day, with instructions to offer only 1 bottle each day, only once per day and to save any formula remaining after the daily ESVFS feeding to return to the study team for measurement. Breastfeeding support was provided on an ongoing basis, and mothers were also taught that they should provide no other food or fluid to enrolled infants other than vitamins, minerals, and medication. For infants randomly assigned to the EBF (control) group, trained study staff provided ongoing breastfeeding support and taught mothers to breastfeed the infant frequently on demand, at least 8 times per day, and to provide no other food or fluid other than vitamins, minerals, and medications.

Measures

Infants and their mothers were enrolled on the day of birth and assessed for baseline characteristics. Data on gestational age were not available because first-trimester ultrasound was rare and pandemic restrictions precluded accurate gestational age assessment in the absence of first-trimester ultrasound.^{40,41} Infants were assessed for outcomes at 4, 14, 30, 60, and 180 days of age.

Infant weight and length were obtained by blinded staff at least twice at each study visit with a Seca 334 scale (Seca Inc, Wandsbek, Germany) and a Seca 416 infant stadiometer, respectively. If the initial 2 weights or 2 lengths differed by >10 g or >0.5 cm, respectively, a third and fourth weight or length were obtained; this occurred for weight and length at 8 and 28 visits, respectively. The interclass correlation coefficient was 1.00 for both weight and length from intra-class correlation coefficient (ICC) (3,k) for test-retest reliability.⁴² Measurements were averaged to obtain final weights and lengths. Infant mid-upper arm circumference (MUAC) and head circumference, maternal height, weight, and MUAC, household dietary intake, infant dietary intake, any infant WHO danger signs or AEs, and hemoglobin were measured by using standard instruments and point-of-care Hemocue (Hemocue, Danaher Inc, Washington, DC).⁴³ For infants receiving ESVFS, formula intake volume was assessed by using prespecified markings (<7.5 mL, 7.5–15 mL, 15–22.5 mL, 22.5–30 mL, 30–45 mL, and 45–59 mL) on the single-serving formula container. Infant deaths were assessed by using the WHO verbal autopsy instrument.

Sample Size and Power

Weight-for-age *z* score (WAZ), weight-for-length *z* score (WLZ), and length-for-age *z* score (LAZ) were calculated by using the WHO Anthro Survey Analyzer⁴⁴; *z* scores <−2 were defined as underweight, wasting, and stunting for WAZ, WLZ, and LAZ, respectively. Conservatively estimating 20% loss to follow-up by 180 days of age, the predetermined sample size of 324 randomized infants was selected to obtain outcomes for 258 infants. This number of outcomes was needed to attain at least 80% power to detect as statistically significant a difference between the treatment groups in WAZ at 30 days of age of 0.20 or larger (SD = 0.6, [Cohen's *d*] *d* = 0.33), a magnitude that would have important public health relevance.

Statistical Analysis

A detailed statistical analysis plan has been published at <https://osf.io/px7ct/>. Linear mixed models were performed by using repeated measures of data collected at all study visits, with an unstructured covariance matrix across time points to test the effect of ESVFS on WAZ, WLZ, and LAZ at 30 days, as well as at other time points, with contrasts between treatment groups at each time

point. A permutation test comparing groups on day 30 was also performed. When the null hypothesis was not rejected ($P < .05$), Bayes Factors⁴⁵ were calculated by comparing models with and without a factor for group on outcomes at day 30.

Primary analyses were performed by using an intention-to-treat approach including all randomized infants. In sensitivity analyses, the primary analyses were repeated by using a per protocol approach comparing infants randomly assigned to ESVFS who consumed more than half the formula provided to infants randomly assigned to EBF who did not receive any formula. Missing data due to dropout was handled by using linear mixed models.^{46,47} Model assumptions were considered satisfied as residuals were approximately symmetric (skewness <2) and with similar variance between groups ($SD^2_1/SD^2_2 = 1.20$).^{48,49} We used linear regression to evaluate the association between the average volume of the supplement consumed by infants randomly assigned to ESVFS and their WAZ change through 30 days of age. χ^2 testing and Fisher's exact tests were used to compare the intervention and control groups with respect to AEs, serious AEs (SAEs), and breastfeeding cessation. All analyses were conducted by using SAS version 9.4 (SAS Institute Inc, Cary, NC).

RESULTS

Of 904 infants initially identified as potentially eligible, PRIMES enrolled 618, of whom 324 (52.1%) met the criteria for randomization and were randomly assigned (Fig 1). Each site randomly assigned 162 infants, of whom 77 (48%) and 72 (44%) were randomly assigned on the day of birth and 85 (52%) and 90 (56%) were randomly assigned at 4 days of age in Guinea-Bissau and Uganda, respectively. The mean ($\pm$ SD) birth weight for those randomly assigned on the day of birth was 2338 ± 135 g, and the mean birth weight for those randomly assigned at 4 days of age was 2620 ± 82 g (Table 1). Among randomly assigned infants, 144 (44%) were male ($P < .05$); of these 144 males, 64 (44%) were randomly assigned to ESVFS ($P = .091$). No other baseline characteristics differed by treatment group.

Infant Feeding Practices

Among infants in the intervention group, the median (interquartile range [IQR]) daily volume of ESVFS feeding ranged from 14.79 (7.39–22.18) mL at 4 days of age to 36.97 (14.79–51.75) mL at 30 days of age and averaged 893 ± 331 mL in total over all feedings (Fig 2). The average volume of the supplement consumed was not associated with WAZ change in the intervention group ($P = .710$). Only 4 (2%) mothers in the control group reported formula use in the first 30 days (Table 2). Nonformula supplementation, including milk, water, sweetened water,

juice, tea, and other liquids, occurred in the first 30 days for 19 infants (12%) in the intervention group and 35 (21%) controls ($P = .02$). Of 304 (94%) infants with follow-up at 180 days of age, 4 infants in the ESVFS group and 3 infants in the control group had stopped breastfeeding ($P = .697$).

Effect of Treatment Assignment

Intention-to-treat and per protocol analyses both revealed that treatment assignment did not significantly affect WAZ, WLZ, or LAZ or the proportion with growth impairment at day 30 based on linear mixed models or permutation tests (Table 3). Bayes Factors of 7.2 to 8.4 revealed “moderate evidence” of the null hypothesis of no group differences at day 30. Tests of effect modification did not identify significant interactions for treatment by site, baseline weight, or sex. The most common AEs in the first 30 days were fever, jaundice, and difficulty breathing, which occurred for 33 (10%), 27 (8%), and 27 (8%) infants, respectively, and did not differ by treatment assignment. The most common AEs in infants >30 days of age were fever, diarrhea, and vomiting, which occurred for 113 (36%), 57 (18%), and 27 (9%) infants, respectively, and did not differ by treatment assignment. In the ESVFS group, 1 infant experienced sudden unexpected infant death at 1 month of age, and another infant died of congenital heart disease diagnosed at 2 months of age, whereas in the EBF group, 2 infants died of fever (one at 7 days of age and one at 4 months of age), and 1 additional infant experienced sudden unexpected infant death at 2 months of age. Overall, in this cohort, the prevalence of growth impairment remained high at 180 days, including stunting (27%), wasting (4%), and being underweight (14%).

Predictors of Infant Growth

As shown in Fig 1, ESVFS intake increased over the first 30 days but remained highly variable between individual infants. Growth outcomes did not differ between infants who consumed >30 mL formula daily and those who consumed <30 mL. Female sex was associated with relative risks of 0.66 (0.42–1.04) and 0.50 (0.34–0.73) for being underweight and stunting, respectively, compared with male infants.

DISCUSSION

In this cohort of otherwise healthy, breastfeeding newborns at risk for growth impairment in Guinea-Bissau and Uganda, the use of ESVFS together with breastfeeding support in the first 30 days did not impact WAZ, WLZ, LAZ, breastfeeding cessation, AEs, or SAEs through 180 days of age. Provided as once-daily feeding immediately after a completed breastfeeding, ESVFS intake was modest, with the average newborn consuming

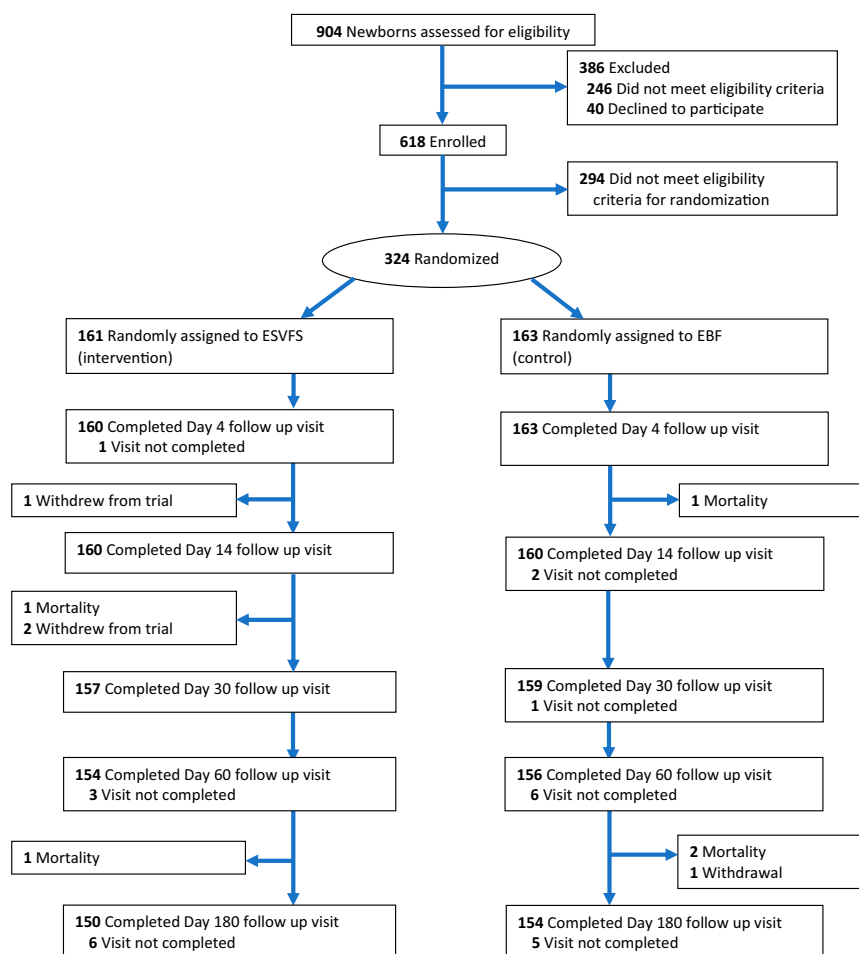


FIGURE 1

Eligibility criteria, sample size, and randomization.

~30 mL/day in addition to breastfeeding. Infants randomly assigned to ESVFS were less likely to receive sweetened water and other liquids during the first 30 days, suggesting that the provision of ESVFS may reduce the use of nonformula, non-breast milk feedings. Taken together, these results suggest that the scientifically quantifiable risks of ESVFS are relatively small and have the potential to be balanced by benefits for some populations.

Designed as a proof-of-concept pilot study with a relatively small sample size of 324 randomly assigned infants, our study reveals 1 method by which early supplemental nutrition can be fed in low-income countries without interfering with ongoing breastfeeding. A technique such as ESVFS could potentially be adapted to allow infants in LMIC access to nutritional support when initial growth during exclusive maternal breastfeeding is inadequate. Unique elements of the ESVFS method used in this trial were the small volume of formula supplement provided, the instruction to supplement formula immediately after completing a breastfeeding, the use of a single-serving, commercially

sterile bottle and single-use nipple along with recommendations for hygienic assembly, and the limited time course for ESVFS supplementation. Using this unique intervention, neither efficacy nor harm was demonstrated in this group of at-risk newborns. It is unknown whether more frequent use or higher volumes of ESVFS, in addition to breastfeeding, might demonstrate efficacy for this population, or whether larger sample sizes or the use of reconstituted powdered infant formula might demonstrate harm. Importantly, it is also unknown whether ESVFS impacts growth for infants whose characteristics differ from PRIMES eligibility criteria. Effect size estimates from this study can be used to inform further research to explore these questions, as well as to determine the effect on the growth of other approaches, including specialized breastfeeding support, donor milk, and maternal milk fortification. In this way, it may be possible to learn how infants in LMIC can attain the same healthy growth as infants in HIC.

This study has important limitations. First, the daily formula intake of infants randomly assigned to ESVFS

TABLE 1 Baseline Characteristics of Randomized Cohort						
	Guinea-Bissau (<i>n</i> = 162)		Uganda (<i>n</i> = 162)		Total Randomized Cohort (<i>n</i> = 324)	
	ESVFS (<i>n</i> = 80)	EBF (<i>n</i> = 82)	ESVFS (<i>n</i> = 81)	EBF (<i>n</i> = 81)	ESVFS (<i>n</i> = 161)	EBF (<i>n</i> = 163)
Infant characteristics on the day of birth						
Wt (g), mean (SD)	2486.69 (166.61)	2493.05 (171.43)	2482.89 (181.94)	2500.33 (194.99)	2484.78 (173.96)	2496.67 (182.98)
WAZ, mean (SD)	−1.85 (0.43)	−1.86 (0.44)	−1.84 (0.5)	−1.8 (0.5)	−1.84 (0.46)	−1.83 (0.47)
Low birth wt (<2500 g), <i>n</i> (%)	38 (47.50)	39 (47.56)	36 (44.44)	36 (44.44)	74 (45.96)	75 (46.01)
Birth length (cm), mean (SD)	45.75 (1.52)	45.88 (1.55)	46.22 (1.32)	46.22 (1.58)	45.99 (1.44)	46.05 (1.57)
LAZ, mean (SD)	−1.98 (0.82)	−1.97 (0.83)	−1.7 (0.74)	−1.71 (0.83)	−1.84 (0.79)	−1.84 (0.84)
WLZ, mean (SD)	−0.77 (0.69)	−0.77 (0.78)	−0.98 (0.65)	−0.91 (0.77)	−0.89 (0.67)	−0.84 (0.78)
Head circumference (cm), mean (SD)	32.15 (1.1)	32.33 (1)	33.02 (1.09)	32.95 (1.11)	32.59 (1.17)	32.64 (1.1)
MUAC (cm), mean (SD)	9 (0.58)	8.99 (0.5)	9.33 (0.57)	9.39 (0.49)	9.16 (0.6)	9.19 (0.54)
Vaginal delivery, <i>n</i> (%)	80 (100.00)	82 (100.00)	73 (90.12)	70 (86.42)	153 (95.03)	152 (93.25)
Female sex, <i>n</i> (%)	44 (55.00)	33 (40.24)	53 (65.43)	50 (61.73)	97 (60.25)	83 (50.92)
Maternal characteristics on the day of birth						
Age (y), median (IQR)	24 (20, 29.5)	25 (22, 30)	22 (20, 28)	23 (20, 26)	22 (20, 28)	24 (21, 28)
Height (cm), mean (SD)	160.04 (5.68)	158.76 (5.97)	159.42 (8.16)	156.96 (8.31)	159.73 (7.02)	157.87 (7.26)
BMI, mean (SD)	24.37 (4.76)	24.28 (4.54)	24.69 (4.72)	24.49 (4.43)	24.53 (4.73)	24.38 (4.48)
MUAC (cm), mean (SD)	26.54 (4.17)	26.6 (3.72)	26.72 (3.94)	26.07 (3.12)	26.63 (4.04)	26.34 (3.44)
MUAC <25 cm, <i>n</i> (%)	34 (42.50)	27 (32.93)	27 (33.33)	34 (41.98)	61 (37.89)	61 (37.42)
Primiparous, <i>n</i> (%)	41 (51.25)	43 (52.44)	41 (50.62)	45 (55.56)	82 (50.93)	88 (53.99)
Years of school, median (IQR)	7 (4, 11)	9 (6, 11)	10 (7, 11)	11 (7, 11)	9 (7, 11)	9 (7, 11)
Years of school <7, <i>n</i> (%)	27 (33.75)	24 (29.27)	4 (4.94)	9 (11.11)	31 (19.25)	33 (20.25)
Marital status, <i>n</i> (%)						
Married, live with husband	36 (45.00)	40 (48.78)	47 (58.02)	46 (56.79)	83 (51.55)	86 (52.76)
Married, live with parents	5 (6.25)	7 (8.54)	1 (1.23)	4 (4.94)	6 (3.73)	11 (6.75)
Not married, live with man	16 (20.00)	14 (17.07)	28 (34.57)	25 (30.86)	44 (27.33)	39 (23.93)
Divorced	0 (0.00)	0 (0.00)	0 (0.00)	1 (1.23)	0 (0.00)	1 (0.61)
Other	23 (28.75)	21 (25.61)	5 (6.17)	5 (6.17)	28 (17.39)	26 (15.95)
Household size, mean (SD)	7.86 (4.86)	10.17 (7.43)	4.46 (2.02)	4.43 (1.98)	6.15 (4.08)	7.32 (6.15)
Place of work, <i>n</i> (%)						
Sleep and work at home	75 (93.75)	76 (92.68)	40 (49.38)	35 (43.21)	115 (71.43)	111 (68.10)
Sleep away for work	0 (0.00)	0 (0.00)	1 (1.23)	0 (0.00)	1 (0.62)	0 (0.00)
Sleep home, work away	5 (6.25)	6 (7.32)	20 (24.69)	19 (23.46)	25 (15.53)	25 (15.34)
Other	0 (0.00)	0 (0.00)	20 (24.69)	27 (33.33)	20 (12.42)	27 (16.56)
Hemoglobin concentration assessed on Day 4						
Infant hemoglobin <12 mg/dL, <i>n</i> (%)	0 (0.00)	0 (0.00)	3 (3.85)	2 (2.47)	3 (1.92)	2 (1.23)
Maternal hemoglobin <10 mg/dL, <i>n</i> (%)	22 (28.21)	20 (24.39)	5 (6.41)	5 (6.17)	27 (17.31)	25 (15.34)
Strata of randomization, <i>n</i> (%)						
Day 0 2000–2249 g	8 (10.00)	8 (9.76)	10 (12.66)	10 (12.35)	18 (11.32)	18 (11.04)
Day 0 2250–2499 g	30 (37.50)	31 (37.80)	26 (32.91)	26 (32.10)	56 (35.22)	57 (34.97)
Day 4 2250–2424 g	4 (5.00)	5 (6.10)	4 (5.06)	5 (6.17)	8 (5.03)	10 (6.13)
Day 4 2425–2599 g	38 (47.50)	38 (46.34)	39 (49.37)	40 (49.38)	77 (48.43)	78 (47.85)

was ~30 mL/day. This small volume may account for the lack of demonstrated efficacy, the lack of detrimental impact, or both. To improve scientific understanding of the relationship between dietary intake and growth and AEs, the authors of future studies in this area should continue to carefully and frequently assess both growth and adverse outcomes. Second, given that actual intake of ESVFS was directly observed by study staff only for the first 4 ESVFS feedings and that subsequent intake was measured from amounts remaining

in bottles when returned by mothers, some recorded formula intake might represent the diversion of formula to non-study uses. However, few bottles were fully consumed, suggesting that the extent of diversion was not large. Third, because reliable data on gestational age was not available, we were not able to assess the impact of prematurity on study outcomes. Although this limited our ability to assess differences between infants born premature and infants born term, it may improve the generalizability of our results

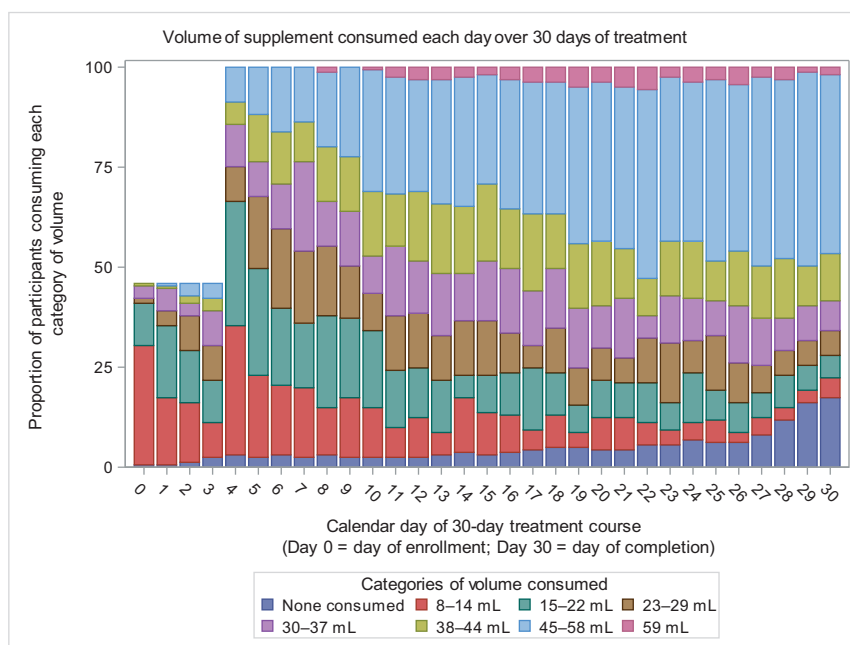


FIGURE 2
ESVFS during the 30-day intervention.

to clinical settings without reliable gestational age estimation. Fourth, we did not study breast milk intake volumes during the period of formula supplementation. Thus, although mothers were taught to breastfeed immediately before providing formula and were encouraged to continue each breastfeeding until the infant discontinued the feeding, we are unable to report whether ESVFS displaced the intake of maternal breast milk in this trial. Emerging technologies for measuring milk intake might facilitate such measurement for future projects.⁵⁰ Fifth, the carefully managed, structured formula supplementation provided in this trial might be difficult to duplicate outside of study conditions.

This limitation reduces the generalizability of our study results. Sixth, although the study findings did not reveal an effect of treatment assignment on AEs, it is important to consider that, because of a limited sample size, the available power was not sufficient to ensure that there were no clinically meaningful risk differences between the groups. Although risk can be calculated scientifically, no intervention is ever entirely free of risk, so the safety of any intervention is a social or policy judgment made in the context of current knowledge regarding risks and benefits.⁵¹ Risk estimates from this study may inform such decisions in the future.

TABLE 2 Infant Dietary Intake by Treatment Assignment						
	0–30 Days of Age			61–180 Days of Age		
	ESVFS (n = 161)	EBF (n = 163)	P*	ESVFS (n = 150)	EBF (n = 154)	P*
	n (%)	n (%)		n (%)	n (%)	
Breast milk	161 (100.00)	163 (100.00)	N/A	146 (97.33)	151 (98.05)	.676
Infant formula	157 ^a (100.00)	4 (2.45)	<.001	15 (10.00)	22 (14.29)	.253
Milk	1 (0.62)	4 (2.45)	.371	41 (27.33)	35 (22.73)	.354
Plain water	9 (5.59)	6 (3.68)	.414	68 (45.33)	80 (51.95)	.249
Sweetened/flavored water	3 (1.86)	8 (4.91)	.219	2 (1.33)	10 (6.49)	.035
Fruit juice	0 (0.00)	3 (1.84)	.248	11 (7.33)	17 (11.04)	.264
Tea/infusion	1 (0.62)	1 (0.61)	0.999	13 (8.67)	13 (8.44)	.944
Other liquids	6 (3.73)	15 (9.20)	.045	6 (4.00)	7 (4.55)	.814
Other (semi) solid food	0 (0.00)	0 (0.00)	N/A	68 (45.33)	81 (52.60)	.205
Any nonformula supplementations	19 (11.80)	35 (21.47)	.020	121 (80.67)	125 (81.17)	.911

* P values are from χ^2 test or Fisher's exact test, as appropriate.
^a 3 withdrawals and 1 infant mortality.

TABLE 3 Effect of Treatment Assignment on Growth Outcomes							
Day	Sample Size	ESVFS (n = 161), Mean (SE)	EBF (n = 163) Mean (SE)	Difference Mean (SE)	P*	P**	Bayes Factor
WAZ							
0	324	−2.03 (0.02)	−2 (0.02)	−0.03 (0.02)	.146		
4	320	−2.22 (0.03)	−2.22 (0.02)	0 (0.03)	.926		
14	317	−1.96 (0.04)	−1.97 (0.04)	0.02 (0.06)	.770		
30	318	−1.74 (0.05)	−1.75 (0.05)	0.01 (0.07)	.895	.668	7.84
60	308	−1.41 (0.07)	−1.43 (0.07)	0.02 (0.09)	.825		
180	302	−1.09 (0.08)	−1.15 (0.08)	0.06 (0.12)	.632		
LAZ							
0	324	−2.06 (0.06)	−2.02 (0.06)	−0.04 (0.08)	.597		
4	319	−2.2 (0.06)	−2.13 (0.06)	−0.07 (0.07)	.297		
14	315	−2.25 (0.07)	−2.17 (0.07)	−0.09 (0.09)	.303		
30	315	−2.05 (0.08)	−2.01 (0.07)	−0.04 (0.1)	.686	.970	7.23
60	308	−1.82 (0.08)	−1.81 (0.08)	−0.01 (0.1)	.901		
180	302	−1.55 (0.09)	−1.65 (0.09)	0.1 (0.12)	.397		
WLZ							
0	250	−0.95 (0.07)	−0.93 (0.07)	−0.02 (0.08)	.829		
4	276	−1.4 (0.07)	−1.48 (0.07)	0.08 (0.09)	.367		
14	307	−0.75 (0.09)	−0.84 (0.09)	0.09 (0.11)	.445		
30	314	−0.12 (0.1)	−0.16 (0.1)	0.04 (0.13)	.758	.778	8.36
60	307	0.23 (0.11)	0.22 (0.11)	0.01 (0.14)	.919		
180	301	−0.09 (0.1)	−0.1 (0.1)	0.01 (0.13)	.918		

* P values are from linear mixed models with contrasts between groups at each time point.

** P values are from a permutation test on day 30.

CONCLUSIONS

In this pilot study of breastfeeding newborns weighing 2000–2499 g at birth or <2600 g at 4 days of age, once-daily ESVFS for 30 days did not alter growth or cause breastfeeding cessation or AEs. Whether different eligibility criteria or more frequent supplement administration might increase growth or affect AEs is unknown. When testing any future interventions, outcomes, including AEs, SAEs, and breastfeeding cessation, should continue to be carefully assessed. In this way, it may be possible to identify a safe, feasible intervention effective at reducing the risk of newborn growth impairment and improving health and development for this important and vulnerable population.

ACKNOWLEDGMENTS

We are grateful for the participation of the infants and their families, the clinical staff at the birthing facilities and clinics, and the team of research nurses and enumerators in Guinea-Bissau and Uganda.

ABBREVIATIONS

AE: adverse event
EBF: exclusive breastfeeding
ESVFS: early small-volume formula supplementation
HIC: high-income countries
IQR: interquartile range
LAZ: length-for-age z score
LMIC: low- and middle-income countries
MUAC: mid-upper arm circumference
PRIMES: Preventing Infant Malnutrition with Early Supplementation
PROBIT: Promotion of Breastfeeding Intervention
SAE: serious adverse event
WAZ: weight-for-age z score
WHO: World Health Organization
WLZ: weight-for-length z score

reviewed and revised the manuscript; and all authors approved the final manuscript as submitted and agree to be accountable for all aspects of the work.

This trial has been registered at www.clinicaltrials.gov (identifier NCT04704076).

Deidentified data and SAS code needed to reproduce the results in this paper will be shared publicly along with a data dictionary at <https://osf.io/px7ct/> on manuscript publication.

DOI: <https://doi.org/10.1542/peds.2023-062228>

Accepted for publication Sep 15, 2023

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PEDIATRICS (ISSN Numbers: Print, 0031-4005; Online, 1098-4275).

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FUNDING: This project was funded by the Bill & Melinda Gates Foundation (OPP1203042). The funder did not participate in the work.

CONFLICT OF INTEREST DISCLOSURES: All authors disclose research funding from the Bill & Melinda Gates Foundation. Dr Flaherman also discloses research funding from the National Heart, Blood and Lung Institute, the Robert Woods Johnson Foundation, the California Department of Health Care Services, the Mt. Zion Health Fund, the Centers for Disease Control Foundation, and BiliCam LLC. Ms Dickinson also discloses research funding from Evolve Biosystems and from Reckitt Benckiser. Dr Allison also discloses funding from Amin Talati Wasserman for KSF Acquisition Corp (Glanbia); Kaleido Biosciences; Law Offices of Ronald Marron; Novo Nordisk Fonden; Sports Research Corp; the United States Department of Agriculture; and WW Inc.

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